

Holly Bender

PROFESSIONAL SUMMARY

Senior Data/AI/ML professional, 13 years in. Track record of taking ambiguous problems through design, delivery and the unglamorous operational work that follows. I mentor deliberately and write things down. Core stack: NumPy, Databricks, BigQuery.

ACADEMIC HISTORY

Master, Biology

Milbrook Institute of Technology | 2010–2011

Dean's List (2 years)

Thesis: "Sequence alignment heuristics for large genomes"

Bachelor, Biology

KTH Royal Institute of Technology | 2006–2009

EMPLOYMENT HISTORY

Beacon Interactive — AI Researcher

2015–2026

- Lifted recommendation click-through by 12% through feature engineering and retraining cadence in Pandas.
- Established offline/online evaluation parity, catching 15 silent regressions before release.
- Ran A/B tests across 10 model variants; the winning variant added an estimated \$100k in annual revenue.
- Built an end-to-end training pipeline on NumPy, cutting retraining from days to 8 hours.

Cobalt Systems — Data Scientist

2013–2015

- Cut inference cost by 20% through quantisation, batching and autoscaling on Datadog.
- Migrated ad-hoc notebooks into versioned Spark pipelines with reproducible runs and tracked experiments.
- Wrote the team's model documentation and review standard, now required before any production launch.

Lumen Works — AI Researcher

2012–2013

- Productionised demand forecasting models that reduced stock-outs by 35% across 5 regions.
- Translated ambiguous business questions into a modelling roadmap with product and finance stakeholders.

TECHNICAL PROFICIENCIES

NumPy, Databricks, BigQuery, Snowflake, LangChain, Airflow, Spark, Pandas, Kubernetes, CI/CD, Selenium, Datadog

SELECTED PROJECTS

Charity chatbot

Built a NumPy assistant for a local charity to answer volunteer questions; retired it once they moved to a supported platform.

Recipe search engine

Full-text search over a scraped recipe corpus using Spark; learned more about ranking than about cooking.

PROFESSIONAL CERTIFICATIONS

Professional Data Engineer — Google Cloud, 2021

Google Analytics Certification — Google, 2025

RESEARCH OUTPUT

"Sequence alignment heuristics for large genomes" - co-authored conference paper, 2013.

VOLUNTEERING

STEM ambassador running school workshops twice a term.